

Program : Diploma in Textile Technology	
Course Code : 4062	Course Title: Fabric Manufacture II
Semester : 4	Credits: 4
Course Category: Program core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- Familiarize the students with auxiliary motions in power loom and evaluate production and efficiency of weaving machines.
- To impart the knowledge of various shedding mechanisms, multiple box looms, terry motion and automatic looms.
- To illustrate the features of shuttle less looms.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Knowledge of basic Mathematics		Mathematics I	1
Knowledge on basic fabric manufacture		Fabric manufacture I	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the auxiliary motion and determine loom parameters.	14	Applying
CO2	Illustrate the dobby and jacquard shedding mechanism.	15	Understanding
CO3	Illustrate the Multiple box, terry motion and automatic looms	14	Understanding
CO4	Familiarize on advanced fabric manufacture.	15	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the auxiliary motion and determine loom parameters		
M1.01	Know the auxiliary motions in loom	4	Understanding
M1.02	Calculate the speed, production & efficiency calculation of various weaving machines.	10	Applying
Contents: Auxiliary motions - Side weft fork and centre weft fork motions-working of side weft fork motion - objectives and types of warp stop motions - working of a mechanical warp stop motion used in 'Sakamoto' looms - objectives and types of warp protector motions - working of loose reed motion - the working of fast reed motion - the objectives and working of brake motion - the objectives and general description of check strap, Temples - Calculate the production and efficiency of power looms - Evaluate the times required for the exhaustion of a weavers beam - Evaluate the time required for the exhaustion of a pirn, cone etc - Estimate the number of warping machines required to feed the given number of looms - Find out the quantity of warp and weft required to weave a particular fabric. Find out the cost of different types of fabrics.			
CO2	Illustrate the dobby and jacquard shedding mechanism		
M2.01	Understand the dobby and Jacquard shedding mechanism	3	Understanding
M2.02	Understand the various type of dobbies	4	Understanding
M2.03	Understand the various types of jacquard	4	Understanding
M2.04	Understand the preparation for jacquard weaving	4	Understanding
	Series Test- I	1	
Contents: Various classification of dobby shedding mechanisms - dobby designs - advantages of double jack dobby over single jack dobby-working of Climax dobby - method of pegging of lags for L.H and R.H dobbies - use of Cross Border dobby-working of Ruti Cam dobby			

- working of Knowles positive open shed dobby - functions of heald leveling device and steadying mechanism in dobby - Basic principle of Jacquard shedding - various classifications of Jacquards - Working of Single lift Jacquards, Working principle of Double Lift Single Cylinder Jacquards, Double Lift double Cylinder Jacquards, Cross border Jacquard - harness mounting like London and Norwich systems - define harness ties and various types of harness ties - count of comber board - First hook and First needle of the jacquard - working of Piano card cutting machine - card punching and card lacing-use of special Jacquards and the working of Pressure Harnes.

CO3	Illustrate the dobby and jacquard shedding mechanism		
M3.01	Know about multiple box motion.	4	Understanding
M3.02	Understand the terry motion.	5	Understanding
M3.03	Know about the automatic looms.	5	Understanding

Contents:

Use of multiple box looms -Type of box motions -. Working of Eccle's Drop box motion-Study of card saving mechanism-Arrangement of pattern cards for various picking patterns-Study of various positions of stud and pin when different boxes are in line with the slay race- Working of Non- Skip type circular box motion- Object and classification of Terry motions-Working of Dug dale's Terry motion-Working of Holden's Terry motion - Features and classification of automatic looms-Advantages of automatic looms- Working of midget weft feeler mechanism- Shuttle protector –functions and uses. Different types of Thread cutters and their uses- Working of automatic weft replenishment motion with respect to cop changing looms -Features of shuttle changing looms.

CO4	Familiarize on advanced fabric manufacture.		
M4.01	Know the working of Sulzer shuttle less weaving	5	Understanding
M4.02	Understand the working of Rapier loom	5	Understanding
M4.03	Understand the working of various jet looms	5	Understanding
	Series Test - II	1	

Contents:

Mention the types of shuttle less looms - Mention the advantages of shuttle less weaving machines - Warp and weft preparation for shuttle less weaving - Mention the various stages of weft insertion in Sulzer shuttle less weaving machine - Explain the Torsion bar picking mechanism - Explain the working of cam beat - up - Introduction to Rapier shuttle less weaving - Various classifications of Rapier shuttle less looms - Various stages of weft insertion in rapier looms - Features of Air Jet looms - The passage of warp yarn on Maxbo Air Jet loom - Functions weft insertion elements used in air jet looms - Various stages of weft insertion on Maxbo air jet looms - Features of water Jet looms - Stages of weft insertion in water jet loom - Working of weft supply system on water jet looms - Mention the merits, demerits of water jet looms - Features of different types of selvages used in shuttle less weaving.

Text / Reference:

T/R	Book Title/Author
R1	Thomas W.Fox - The Mechanism of Weaving - Universal Publishing corp., Mumbai-2
R2	NCUTE - Woven Fabric Production II – NCUTE New Delhi
R3	N.N.Banerjee - Weaving Mechanism Vol.I - Smt Tandra banerjee & Smt Apurba banerjee, West Bengal
R4	Prof.J.C.Chakravarthi - Mechanism of Weaving Vol.I & II - Prof. C. Chakravarthi West Bengal
R5	R.Sengupta- Weaving Calculations - D.B.Taraporevala sons & co Ltd., Mumbai

Online resources:

Sl.No	Website Link
1	http://freetexbooks.blogspot.in/2011/10/handbook-of-weaving.html
2	http://www.nptel.ac.in/courses
3	http://textilelearner.blogspot.in
4	https://textilestudycenter.com